SCM LAB REPORT

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1. Introduction

What is Git?

Git is a version control system used to track changes in source code during software development. It helps multiple developers collaborate efficiently.

What is Version Control?

Version control is a system that records changes to files over time so that specific versions can be recalled later.

What is GitHub?

GitHub is a cloud-based platform used to host Git repositories and enable collaboration among developers.

Difference Between Git and GitHub

- Git is a tool, GitHub is a platform
- Git works locally, GitHub works online
- Git tracks changes, GitHub stores repositories
- Git doesn't require internet, GitHub does

2. Basic git commands

git init

Description: Initializes a new Git repository

Git status

Description: Shows current status of file

```
MINGW64:/c/Users/Acer/scm-lab

Acer@DESKTOP-58S1OUS MINGW64 ~
$ mkdir scm-lab

Acer@DESKTOP-58S1OUS MINGW64 ~
$ cd scm-lab

Acer@DESKTOP-58S1OUS MINGW64 ~/scm-lab
$ git init
Initialized empty Git repository in C:/Users/Acer/scm-lab/.git/

Acer@DESKTOP-58S1OUS MINGW64 ~/scm-lab (master)
$ touch file1.txt

Acer@DESKTOP-58S1OUS MINGW64 ~/scm-lab (master)
$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        file1.txt

nothing added to commit but untracked files present (use "git add" to track)

Acer@DESKTOP-58S1OUS MINGW64 ~/scm-lab (master)
$ git add .

Acer@DESKTOP-58S1OUS MINGW64 ~/scm-lab (master)
$ git commit -m "first commit"
[master (root-commit) b05658d] first commit
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 file1.txt
```